Pan Peter B.Sc.

Model Predictive Control for Autonomous Vehicles: A Simulation-Based Approach

Master's Thesis

to achieve the university degree of

Master of Science
Master's degree programme: Electrical Engineering

submitted to
Graz University of Technology

Supervisor
Univ.-Prof. Dipl-Ing. Dr.techn. Markus Reichhartinger

Institute of Automation and Control
Head: Univ.-Prof. Dipl-Ing. Dr.techn. Markus Reichhartinger

Graz, March 2025

Affidavit

I declare that I have authored this thesis independently, that I have not used other than the declared sources/resources, and that I have explicitly indicated all material which has been quoted either literally or by content from the sources used.

During the preparation of this thesis in **ADD YEAR**, the AI tool **e.g. ChatGPT (LLM: GPT-5), developed by OpenAI**, was used to improve language quality and readability. The tool is available at **https://chat.openai.com/**. All generated suggestions were carefully reviewed and revised by the author. The author takes full responsibility for the content of this work.

The text document uploaded to TUGRAZonline is identical to the present master‘s thesis.

Date	Signature

Eidesstattliche Erklärung

Ich erkläre an Eides statt, dass ich die vorliegende Arbeit selbstständig verfasst, andere als die angegebenen Quellen/Hilfsmittel nicht benutzt, und die den benutzten Quellen wörtlich und inhaltlich entnommenen Stellen als solche kenntlich gemacht habe.

Während der Erstellung dieser Arbeit im Jahr **JAHR** wurde das KI-Tool **z. B. ChatGPT (LLM: GPT-5), entwickelt von OpenAI**, eingesetzt, um die sprachliche Qualität und die Lesbarkeit zu verbessern. Das Tool ist unter **https://chat.openai.com/** verfügbar. Alle generierten Vorschläge wurden **vom Autor/von der Autorin** sorgfältig geprüft und überarbeitet. Der Autor übernimmt die volle Verantwortung für den Inhalt dieser Arbeit.

Das in TUGRAZonline hochgeladene Textdokument ist mit der vorliegenden Masterarbeit identisch.

Datum	Unterschrift

Abstract

Brief outlook on the content of this document.

Kurzfassung

Kurze Beschreibung des Inhalts.

Acknowledgements

Personal acknowledgements ...

Contents

1 Introduction **1**
1.1 Background 2
1.1.1 Sub-section 1 2
1.2 Problem Statement 2

2 Methodology **3**
2.1 A table 3
2.2 Some equations 3
2.3 Theorem 5
2.3.1 Theorem 1 5
2.3.2 Theorem 2 with proof 5

List of Tables

2.1 Constants of a system 4

List of Figures

1.1 Short caption text 1
1.2 Short overall caption 1

List of Acronyms

NASA National Aeronautics and Space Administration

Chapter 1

Introduction

Content of the Introduction. The Introduction 1 holds a figure 1.1, a figure 1.2 and cites something [1]. Afterwards it continues with chapter 2. The National Aeronautics and Space Administration (NASA) was founded in 1958. The NASA is known for its space missions. Use the command $\backslash myTODO$ to mark something that needs to be done, e.g., TODO: Add more content here.

Figure 1.1: Long caption text

(a) Caption figure A

(b) Caption figure B

Figure 1.2: Overall caption

1.1 Background

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

1.1.1 Sub-section 1

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

1.2 Problem Statement

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Chapter 2

Methodology

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

2.1 A table

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

2.2 Some equations

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames

Table 2.1: Constants of a system

Parameter	Value	Unit
m	0.19	kg
k	0.2	$\frac{\mathrm{kg}}{\mathrm{s}}$
c	4.8398	$\frac{\mathrm{N}}{\mathrm{m}}$

ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

$$\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}u, \qquad y = \boldsymbol{c}^T\boldsymbol{x} + du.$$

$$\begin{bmatrix} \boldsymbol{x}_{k+1} \\ u_k \end{bmatrix} = \underbrace{\begin{bmatrix} \boldsymbol{A}_d & \boldsymbol{\Gamma}_1(\tau_a,\tau_b) \\ \boldsymbol{0} & 0 \end{bmatrix}}_{=:\boldsymbol{A}_\xi(\tau_a,\tau_b)} \underbrace{\begin{bmatrix} \boldsymbol{x}_k \\ u_{k-1} \end{bmatrix}}_{=:\boldsymbol{\xi}_k} + \underbrace{\begin{bmatrix} \boldsymbol{\Gamma}_0(\tau_a,\tau_b) \\ 1 \end{bmatrix}}_{=:\boldsymbol{b}_\xi(\tau_a,\tau_b)} u_k, \tag{2.1}$$

$$y_k = \begin{bmatrix} \boldsymbol{c}^T & 0 \end{bmatrix} \boldsymbol{\xi}_k,$$

$$\begin{bmatrix} \boldsymbol{x}_{k+1} \\ u_k \\ u_{k-1} \\ \vdots \\ u_{k-\zeta+1} \end{bmatrix} = \underbrace{\begin{bmatrix} \boldsymbol{A}_d & \Gamma_1(\mathrm{Y}) & \dots & \Gamma_{\zeta-1}(\mathrm{Y}) & \Gamma_\zeta(\mathrm{Y}) \\ \boldsymbol{0} & 0 & \dots & 0 & 0 \\ \boldsymbol{0} & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \boldsymbol{0} & 0 & \dots & 1 & 0 \end{bmatrix}}_{=:\boldsymbol{A}_\xi(\mathrm{Y})} \underbrace{\begin{bmatrix} \boldsymbol{x}_k \\ u_{k-1} \\ u_{k-2} \\ \vdots \\ u_{k-\zeta} \end{bmatrix}}_{=:\boldsymbol{\xi}_k}$$

$$+ \underbrace{\begin{bmatrix} \Gamma_0(\mathrm{Y}) \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}}_{=:\boldsymbol{b}_\xi(\mathrm{Y})} u_k, \tag{2.2}$$

$$y_k = \begin{bmatrix} \boldsymbol{c}^T & 0 & 0 & \dots & 0 \end{bmatrix} \boldsymbol{\xi}_k.$$

2.3 Theorem

2.3.1 Theorem 1

Theorem 3.1: Pythagoras' theorem

In a right triangle, the square of the hypotenuse is equal to the sum of the squares of the catheti.

$$a^2 + b^2 = c^2$$

In mathematics, the Pythagorean theorem, also known as Pythagoras' theorem, is a relation in Euclidean geometry among the three sides of a right triangle.

2.3.2 Theorem 2 with proof

Theorem 3.2

There exist two irrational numbers x, y such that x^y is rational.

Proof 3.1

If $x = y = \sqrt{2}$ is an example, then we are done; otherwise $\sqrt{2}^{\sqrt{2}}$ is irrational, in which case taking $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$ gives us:

$$\left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = \sqrt{2}^{\sqrt{2}\sqrt{2}} = \sqrt{2}^{2} = 2.$$

Bibliography

[1] P. Peter, "Latex: A document preparation system," *Software: Practice and Experience*, vol. 23, no. 10, pp. 141–152, 1995.